
SCMRE: A SITUATED CORPUS FOR GENERATING MULTIMODAL REFERRING EXPRESSIONS BY EMBODIED VIRTUAL AGENTS

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ABSTRACT

Generating definite descriptions using both verbal and non-verbal modalities is a critical human capability to indicate objects and locations in collaborative interactions. Modern chatbots, exemplified by OpenAI's ChatGPT, have demonstrated remarkable abilities in generating coherent, context-relevant text. Nonetheless, relying on the text only fails to demonstrate understanding of the *meaning* that connects utterance to communicative intent Bender and Koller (2020). Agent embodiment provides a structure to understand language in context Krishnaswamy and Alalyani (2021). If a particular mode of expression (e.g., language) is inadequately communicative, another mode (e.g., gesture) can be used to disambiguate targets. AI advancements have developed systems that enable humans to interact with computers multimodally Chen et al. (2019); Li et al. (2022); Qi et al. (2020); Shridhar et al. (2020), but embodied interactive virtual agents (IVAs) cannot typically intelligently mix modalities to communicate their intents as humans do. Hence, exploring human referential behaviors is essential for developing IVAs that bidirectionally and symmetrically collaborate with humans. We present **SCMRE**, a situated corpus of multimodal reffering expressions intended for training generative AI systems for multimodal HCI. Our presentation will involve the following contributions. 1) Developing an IVA platform that interprets human multimodal instructions and responds with language and gestures to elicit human references. These include attributive, relational, or historical referring expressions, which may use only deictic references or ensembles of deixis and language. 2) Establishing human-IVA communication, providing 24 participants with 10 scenes, each involving ten equally-sized blocks randomly placed on a table. These interactions produced a dataset of 10,408 samples, each including target objects, distance from the agent, object coordinates, object relations, previous events, utterance, modality(ies), and demonstratives. 194 videos were recorded, spanning over 36 hours. 3) Analyzing SCMRE, revealing a correlation between modalities and the degree of ambiguity, indicating that the utilization of pointing significantly reduces the ambiguity of prompts (p -value $< 2.2e - 16$ using Pearson's Chi-squared test). 4) Proposing a generative model to contextually deploy multimodal references, including language and gesture.

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